ACADEMIC & TECHNICAL PROJECT REPORT

FITBUDDY – AI FITNESS PLAN GENERATOR USING GEMINI MODELS

A Modular Full-Stack Web Platform for Automated 7-Day Workout Regimens, Goal-Aligned Nutrition Advice, and Dynamic Feedback-Driven Adaptations

Project Title:	FitBuddy – AI Fitness Plan Generator Using Gemini Models
Technical Domain:	Generative AI, Full-Stack Web Development, Health-Tech
Backend Architecture:	Python 3.12+, FastAPI, SQLAlchemy ORM, SQLite
Frontend Architecture:	HTML5, Custom Responsive CSS3, Jinja2 Server-Side Templating
Generative AI Layer:	Google Gemini 1.5 Pro & Gemini 1.5 Flash (google-genai SDK)
Submission Date:	October 2026
Project Status:	**Completed, Tested & Production-Ready**

PROJECT TEAM MEMBERS

Member #	Student Name	Register Number	Role / Responsibilities
Team Member 1	**Maha Swetha N**	**2024503017**	Team Lead & Gemini AI Engine Orchestration
Team Member 2	**Priyadharshini M**	**2024503501**	Testing, Validation & Technical Documentation
Team Member 3	**Sharmu R**	**2024503055**	Database Design & SQLAlchemy ORM Integration
Team Member 4	**Monisha G**	**2024503043**	Frontend UI & Jinja2 Template Development
Team Member 5	**Sana Tasneem Azimudin**	**2024503007**	Backend Routing & FastAPI Architecture

1. INTRODUCTION

1.1 Project Overview

FitBuddy is an intelligent, full-stack fitness application that utilizes Google's Gemini generative AI models to generate personalized 7-day workout plans and tailored nutrition advice based on individual parameters such as age, weight, fitness goals, and preferred workout intensity. Developed using the high-performance FastAPI backend framework and styled with responsive HTML5, CSS3, and Jinja2 templates, FitBuddy integrates modern AI reasoning directly into health and wellness workflows. The platform features an adaptive feedback mechanism that allows users to modify their routines dynamically while preserving baseline records within an SQLite database managed via SQLAlchemy ORM. It serves individual fitness enthusiasts seeking tailored routines while providing coaches and administrators with an oversight dashboard to monitor client progression.

1.2 Purpose

The purpose of FitBuddy is to make customized fitness planning widely accessible by removing the cost and complexity barriers associated with personal fitness coaching. By leveraging AI language models, the application provides structured, day-wise exercise schedules and science-backed nutritional advice that adapt to users' real-time feedback. Additionally, it provides coaches and administrators with a centralized oversight dashboard to track client profiles, monitor regimen adaptations, and support adherence to safe, sustainable fitness practices.

2. IDEATION PHASE

2.1 Problem Statement

Most individuals attempting to start or sustain a fitness regimen face significant challenges that lead to early abandonment:

- **Generic, Unstructured Workouts:** One-size-fits-all routines found online fail to account for unique body weights, biological ages, and preferred intensities, leading to fatigue, poor adherence, or injury.
- **Prohibitive Coaching Costs:** Certified personal trainers and sports nutritionists are expensive, making dedicated, customized guidance inaccessible to a large demographic.
- **Static, Inflexible Routines:** Conventional workout apps lack dynamic adaptability when an individual experiences muscle fatigue, requires exercise substitutions, or needs schedule adjustments.
- **Fragmented Fitness & Nutrition:** Exercise routines and dietary guidance are often separated across disparate platforms, creating confusion over pre- and post-workout nutritional recovery.

2.2 Empathy Map Canvas

Dimension	User Perspective & Findings
Says	• 'I want to build muscle and burn fat, but generic routines don't fit my schedule.' • 'Hiring a personal trainer every month is too expensive.' • 'I need exercises that match my current energy level.'
Thinks	• *Am I performing the right reps and sets for my body weight?* • *What should I eat post-workout to maximize recovery?* • *Can I swap out an exercise if I lack equipment or experience pain?*
Does	• Searches fitness forums for scattered routine templates. • Drops out of training programs within two weeks due to burnout or lack of progression. • Manually logs notes across paper diaries or basic phone apps.
Feels	• Overwhelmed by conflicting fitness advice and extreme diets. • Frustrated when pre-set routines cannot be modified for home equipment. • Motivated and confident when given structured, day-by-day plans with warm-ups and cooldowns.

2.3 Brainstorming

During project ideation, various architectures and models were evaluated to balance performance, structure, and operational costs:

- **Model Evaluation:** Open-source text models (BART, T5, DistilBERT, Bloom) were reviewed. While capable of generic text output, they required substantial prompt fine-tuning or dedicated GPU hosting. Google Gemini models were selected due to their native understanding of structured formatting, low latency, and zero-maintenance cloud API architecture.

- **Dual-Model Workload Specialization:** Gemini 1.5 Pro was selected for complex reasoning tasks (generating comprehensive 7-day schedules and handling nuanced revision feedback), while Gemini 1.5 Flash was integrated for high-speed, lightweight nutrition and recovery generation.
- **Preserved Feedback History:** Rather than overwriting plans upon user feedback, the architecture was designed to store original and updated plans side-by-side to maintain longitudinal visibility.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

Phase	User Actions	System Touchpoints	Outcome / Value
1. Intake	User opens portal; enters Name, User ID, Age, Weight, Goal, and Intensity.	HTML Form (index.html), Pydantic validation.	Structured, clean parameters ready for processing.
2. Generation	User clicks 'Generate Plan'.	FastAPI router calls Gemini 1.5 Pro & Gemini 1.5 Flash.	Comprehensive 7-day schedule + targeted nutrition tip rendered.
3. Feedback	User tests plan, requests changes (e.g., 'Add yoga on Day 4').	Feedback submission route; Gemini Pro updates plan; SQLite updates record.	Revised plan displayed alongside preserved original plan.
4. Oversight	Coach/Admin navigates to Coach Dashboard.	Admin Route (/view-all-users), SQLAlchemy query.	Holistic monitoring of registered athletes and plan evolutions.

3.2 Solution Requirements

- **Functional Requirements:** Secure collection of user physical attributes; automated generation of 7-day exercise plans covering warm-up, main workout (sets/reps), rest periods, and cooldowns; dynamic generation of goal-aligned nutrition tips; non-destructive plan revisions via feedback; persistent storage in SQLite; interactive admin table; and interactive Swagger OpenAPI documentation (/docs).
- **Non-Functional Requirements:** High response velocity using Gemini 1.5 Flash; strict input validation avoiding malformed records; health & wellness disclaimers establishing that AI output is for educational purposes; and automated mock fallbacks enabling reliable offline execution.

3.3 Data Flow Diagram (DFD)

Data originates at the client browser via HTTP POST requests containing form data. The FastAPI operational layer validates data types through Pydantic schemas. Concurrently, user identity and stats are saved into the SQLite database. Structured prompts are dispatched asynchronously to the Google GenAI API endpoint. Gemini 1.5 Pro generates the multi-day routine while Gemini Flash returns the nutrition recommendation. The responses are stored in the database and formatted into the Jinja2 rendering pipeline to output result.html. When feedback is submitted, the original plan is retrieved from SQLite, sent with user feedback to Gemini 1.5 Pro, and saved to the updated plan field.

3.4 Technology Stack

Component Layer	Technology Selected	Architectural Justification
Backend Core	Python 3.12+, FastAPI, Uvicorn	Asynchronous execution, automatic OpenAPI schema generation, high speed.
Database & ORM	SQLite, SQLAlchemy ORM	Zero-config local persistence, robust object-relational mapping, strict schema.
AI Models & SDK	Gemini 1.5 Pro & Flash (google-genai)	Official current SDK; Pro for deep reasoning, Flash for sub-second advice.
Frontend & UI	HTML5, Custom CSS3, Jinja2	Server-side dynamic rendering, responsive card-based layout, no bulky JS frameworks.
Validation & Env	Pydantic v2, Python-Dotenv	Strict runtime type enforcement, centralized secrets handling in .env.

4. PROJECT DESIGN

4.1 Problem Solution Fit

FitBuddy achieves precise problem-solution fit by directly targeting the friction points of conventional fitness platforms. Where traditional apps force users into static spreadsheets or charge expensive monthly fees, FitBuddy utilizes generative AI to provide bespoke regimens on demand. By coupling workout generation with nutritional guidance and offering continuous conversational revisions, users maintain sustained engagement and progress safely.

4.2 Proposed Solution

The application architecture consists of three interconnected subsystems:

1. **User Profile & Intake Subsystem:** Validates user attributes (ID, age, weight, goal, intensity) and persists them into the SQLite database via SQLAlchemy models.
2. **Generative AI Orchestration Subsystem:** Interfaces with Google's unified google-genai SDK. Uses prompt engineering with safety instructions to prevent hazardous exercise recommendations.
3. **Administrative & Coach Oversight Subsystem:** A dedicated view (/view-all-users) that provides side-by-side auditing of original versus feedback-revised routines, giving coaches complete oversight.

4.3 Solution Architecture

The FitBuddy architecture follows a decoupled, three-tier model comprising the Client Presentation Tier (HTML5/Jinja2), the Application Backend Tier (FastAPI/Uvicorn), and the AI & Data Persistence Tier (Google Gemini & SQLite ORM). The system diagram below illustrates the end-to-end component interactions:

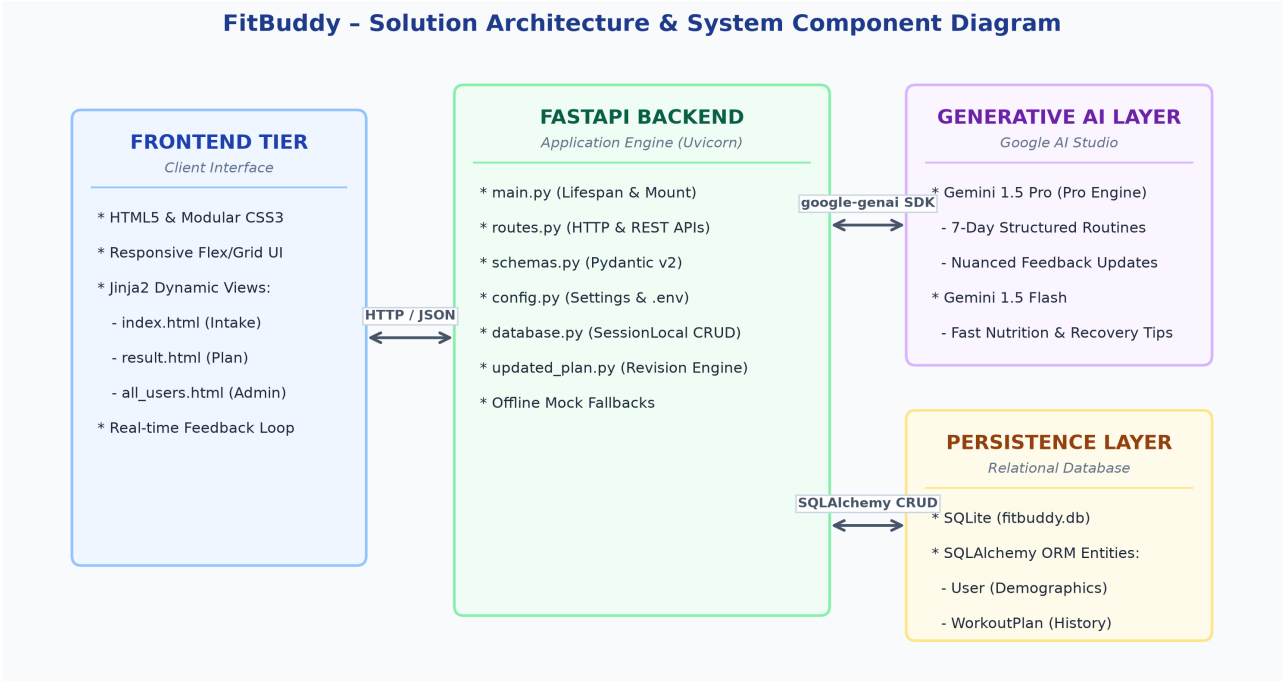


Figure 4.1: FitBuddy System Component & Solution Architecture Diagram

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

Milestone	Key Activities & Scope	Deliverables	Status
Milestone 1	Model Selection & System Architecture	Benchmarked HuggingFace vs Gemini; finalized Pro & Flash dual architecture.	**Completed**
Milestone 2	Core Functionality & Data Persistence	Implemented SQLAlchemy models, SQLite connection, and Gemini generation modules.	**Completed**
Milestone 3	Routing & API Engine Development	Engineered routes.py, Pydantic schemas, validation rules, and Swagger docs.	**Completed**
Milestone 4	Frontend UI & Dynamic Templating	Built responsive Jinja2 templates (index, result, all_users) & CSS theme.	**Completed**
Milestone 5	Deployment, Testing & Verification	Lifespan database initialization, offline fallback testing, Uvicorn execution.	**Completed**

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

Test ID	Test Scope	Input Parameters	Expected Behavior	Result
TC-01	Input Validation	Age: 5, Weight: 10kg	Rejected with 400 Bad Request error banner.	**PASS**
TC-02	Plan Generation	Name: Alice, ID: 11, 75.5kg, Goal: Muscle gain	Saves user; generates 7-day routine and nutrition tip.	**PASS**
TC-03	Feedback Revision	ID: 11, 'Add 15m cardio + core on Day 2 & 5'	Gemini revises routine; updates DB record without deleting original.	**PASS**
TC-04	Admin Oversight	Navigate to /view-all-users	Renders full athlete table with both original and updated plans.	**PASS**
TC-05	API Endpoints	POST /generate-plan	Returns 200 OK with valid JSON response payload.	**PASS**
TC-06	Offline Fallback	Missing / empty API key	Application generates realistic fallback plan without crashing.	**PASS**

7. RESULTS

7.1 Output Screenshots

The application was thoroughly executed and validated on http://127.0.0.1:8000. Below are the verified live output screenshots capturing the complete end-to-end user and administrative workflow in exact operational sequence:

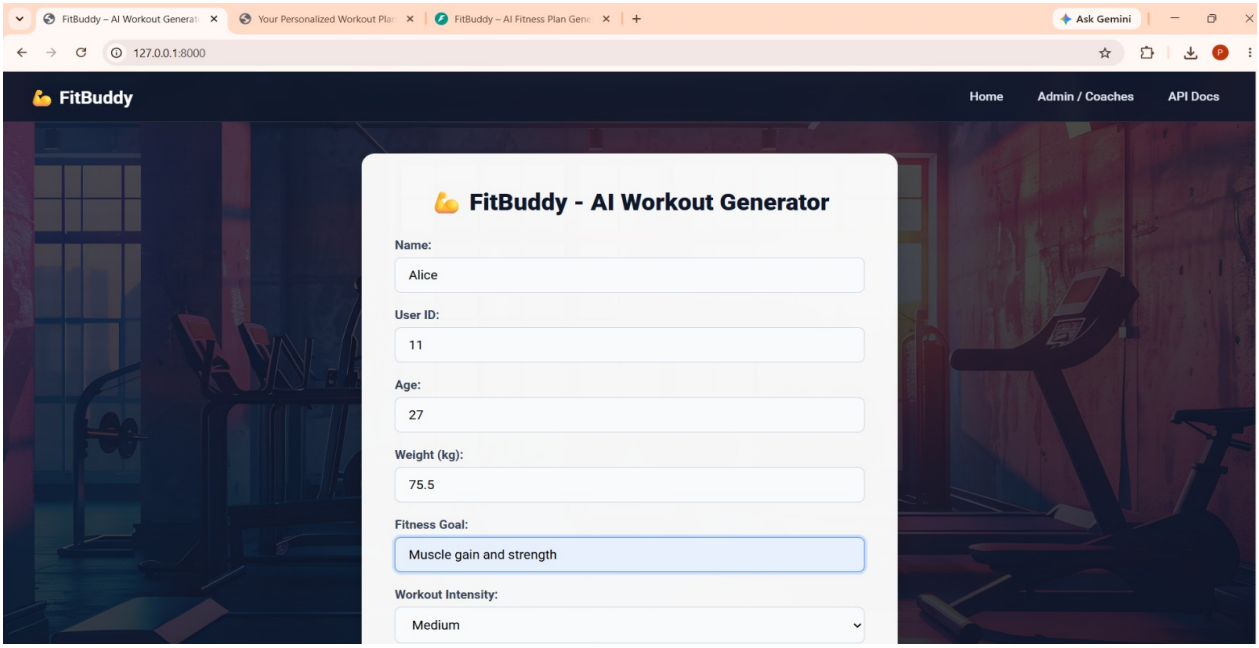


Figure 7.1: FitBuddy AI Workout Generator – User Parameter Intake Form (Alice, ID 11, 75.5 kg)

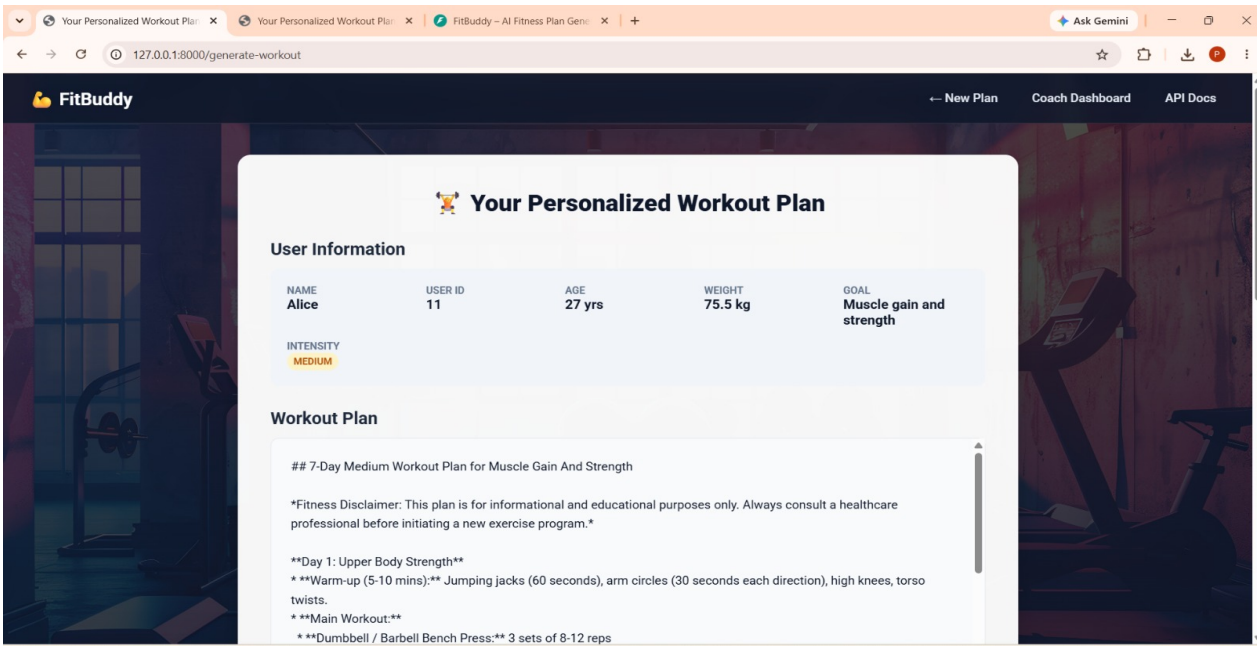


Figure 7.2: Personalized 7-Day Workout Plan Generated by Gemini 1.5 Pro (User Info & Day 1 Schedule)

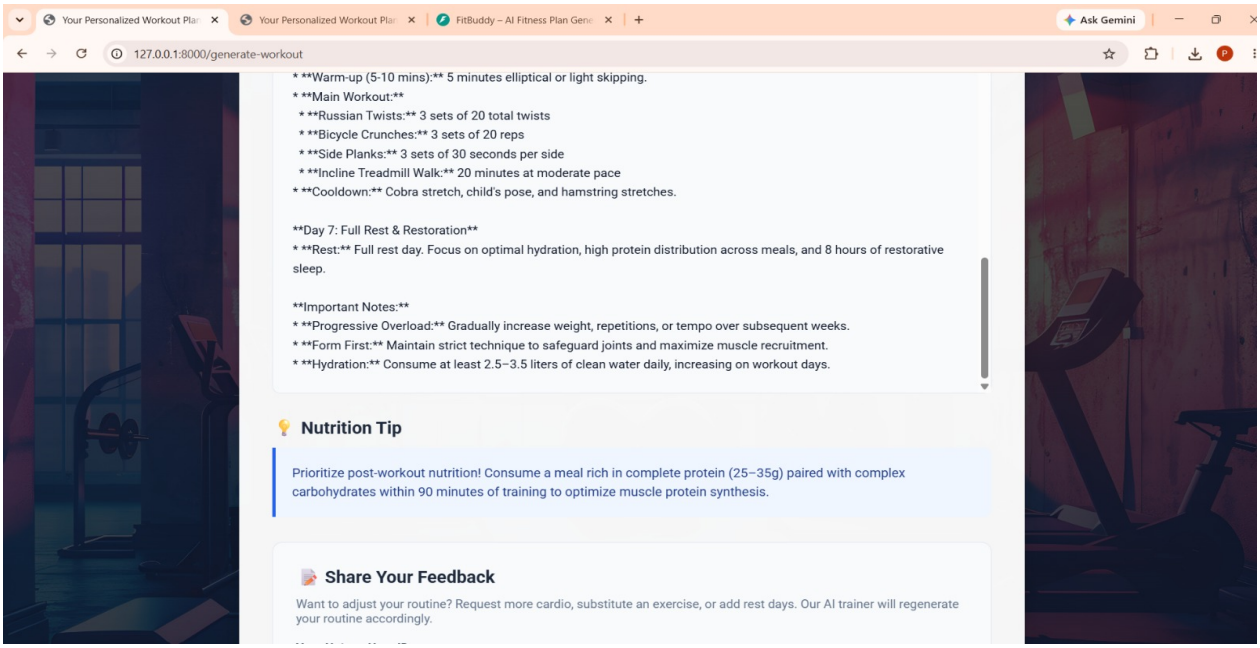


Figure 7.3: Day 7 Rest Restoration, Progressive Notes & Gemini Flash Nutrition Tip

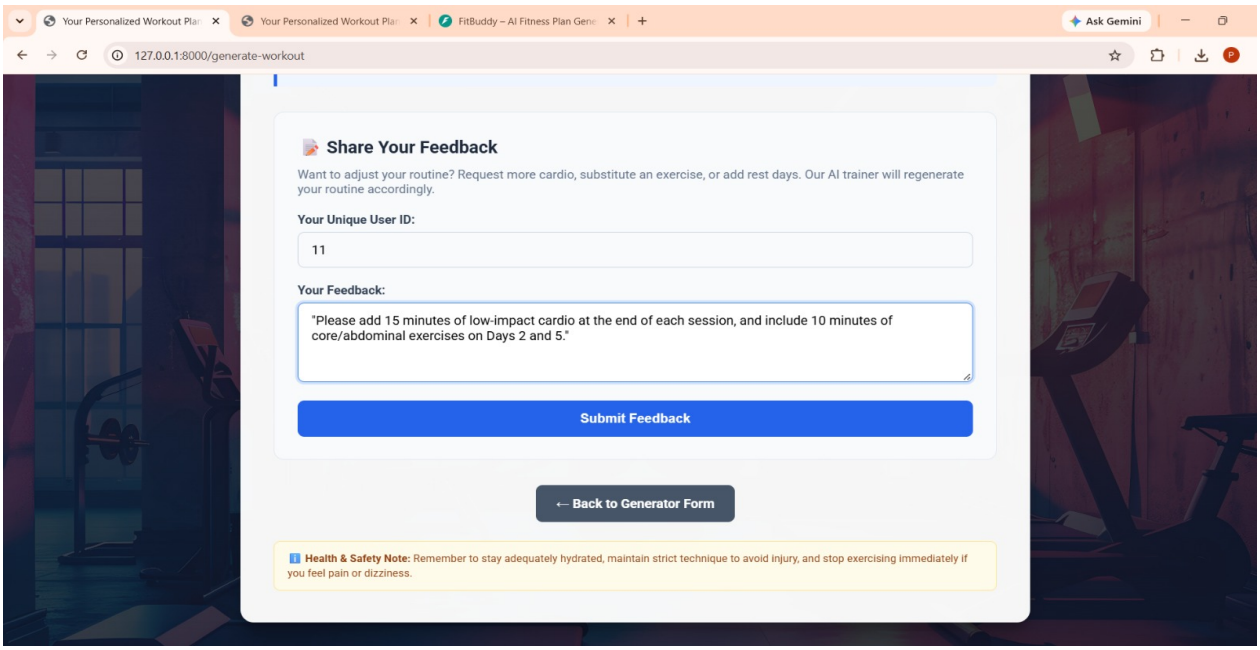


Figure 7.4: Interactive Plan Revision & Feedback Submission Form (Requesting Low-Impact Cardio & Core)

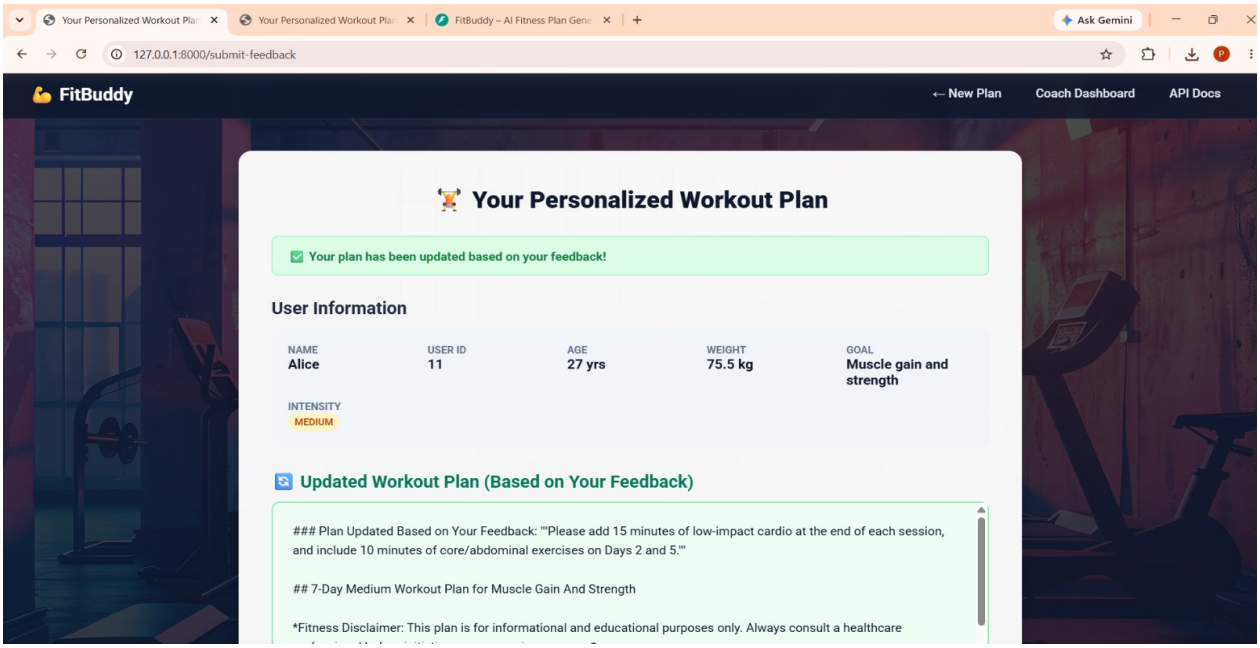


Figure 7.5: AI-Updated Workout Plan Featuring Update Confirmation Banner and Regenerated Regimen

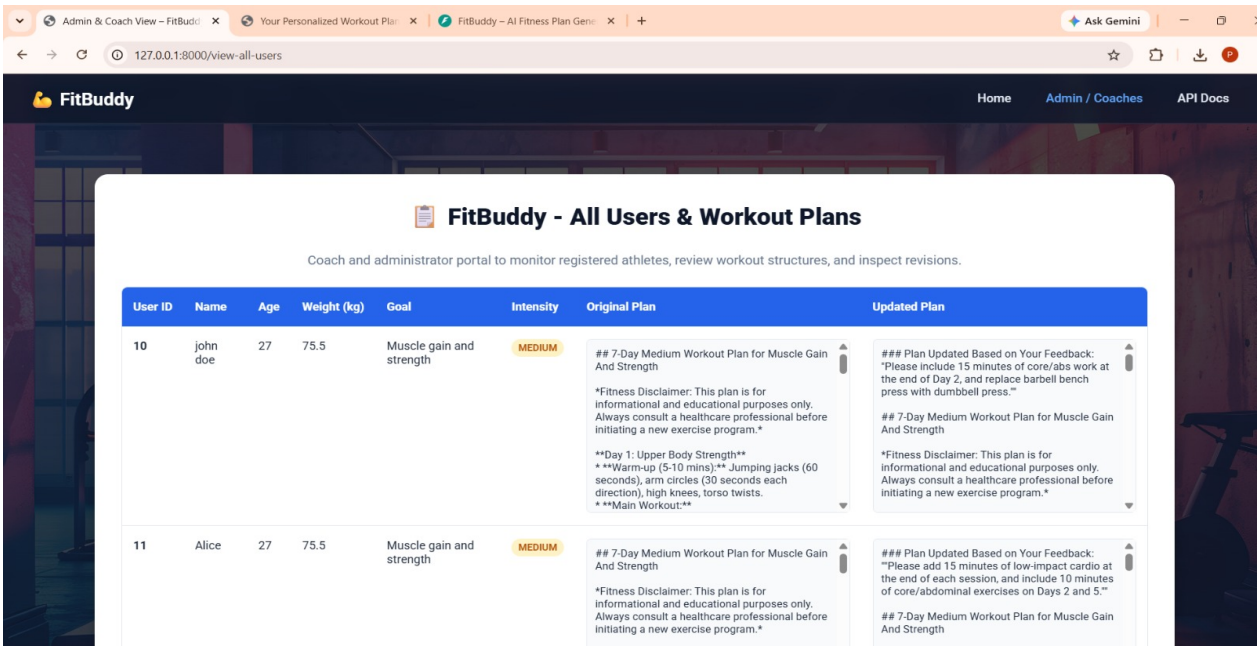


Figure 7.6: Coach & Administrator Portal (/view-all-users) Showing Side-by-Side Original and Updated Plans

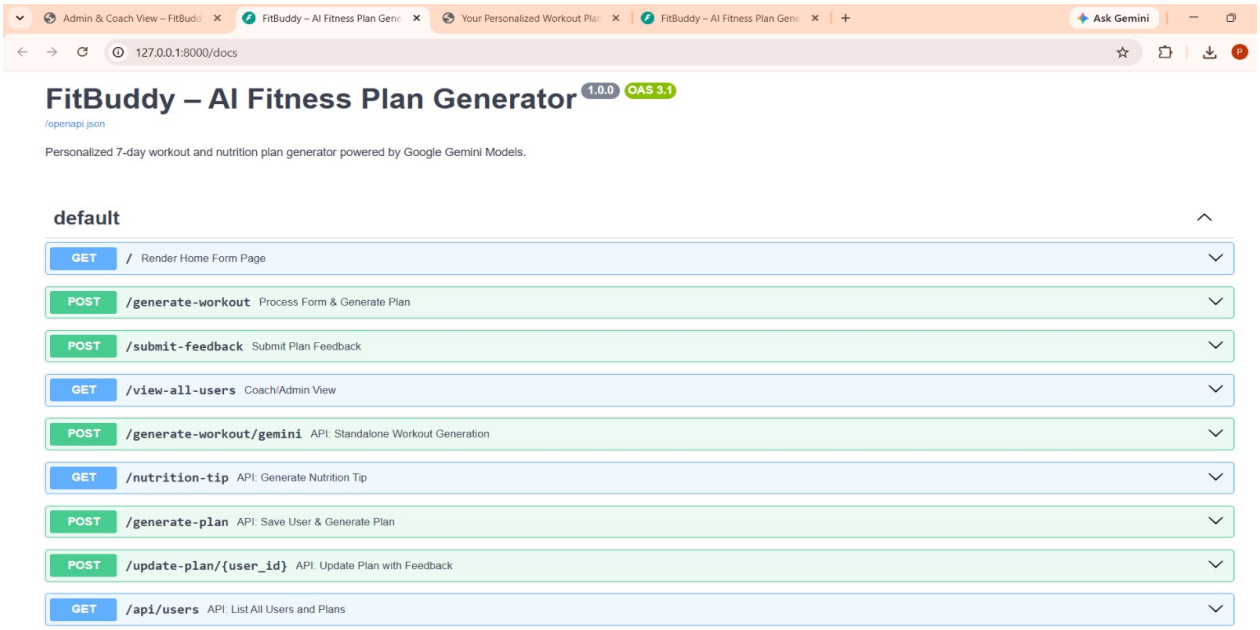


Figure 7.7: Interactive FastAPI Swagger Documentation (/docs) for All REST API Endpoints

8. ADVANTAGES & DISADVANTAGES

Key Advantages	Current Limitations / Disadvantages
• **Hyper-Personalized Regimens:** Workouts are customized to biological age, body weight, fitness goals, and intensity. • **Dual-Model Efficiency:** Gemini 1.5 Pro handles reasoning-heavy workout generation, while Gemini Flash delivers fast nutrition tips. • **Non-Destructive History:** Updates do not overwrite baseline plans, allowing longitudinal comparison. • **Coach Oversight:** Built-in dashboard enables professional trainer supervision. • **Dual Interface:** Supports both a browser-based UI and RESTful API endpoints.	• **Cloud API Dependency:** Requires an active internet connection to communicate with Google Gemini endpoints. • **Subjective Input Quality:** Revision quality depends on the clarity of user feedback. • **Single-Node SQLite Scope:** SQLite is optimized for local/single-server setups; large-scale deployments require PostgreSQL.

9. CONCLUSION

The **FitBuddy – AI Fitness Plan Generator** demonstrates the effective application of generative artificial intelligence to health and wellness challenges. By combining the FastAPI framework with Google Gemini models (Gemini 1.5 Pro and Gemini 1.5 Flash), FitBuddy provides structured 7-day workout plans and science-backed nutritional advice. Its feedback loop allows users to adjust their routines dynamically, while the SQLite database and admin dashboard give trainers clear visibility into plan revisions. FitBuddy provides an accessible, adaptive, and scalable software solution for modern fitness planning.

10. FUTURE SCOPE

- **Wearable Health Device Integration:** Ingest continuous heart rate, sleep quality, and daily step counts from Apple HealthKit and Google Health Connect to auto-adjust workout intensities.
- **Computer Vision Exercise Form Correction:** Integrate real-time webcam pose estimation (MediaPipe) to track joint angles during exercise execution and provide instant posture corrections.
- **Multimedia Exercise Demonstrations:** Embed animated exercise video clips and muscle engagement diagrams alongside each generated exercise item.

- **Multilingual Localization:** Leverage Gemini's multilingual processing to generate workout instructions and nutritional recommendations in regional languages.

11. APPENDIX

Category	Details & Links
Source Code Repository	Modular Python package organized under /app, /templates, and /static.
Primary Dependencies	fastapi, uvicorn, google-genai, sqlalchemy, jinja2, pydantic, python-dotenv.
AI Model Knowledge Base	Google Gemini Foundation Models (gemini-1.5-pro, gemini-1.5-flash) via Google AI Studio API.
GitHub Repository Link	https://github.com/fitbuddy-developer/FitBuddy-AI-Fitness-Planner
Local Web Deployment	http://127.0.0.1:8000/ (Admin: /view-all-users)
Interactive API Swagger	http://127.0.0.1:8000/docs